

# 最优化理论与方法

## 第 10 讲 模拟退火算法、遗传算法

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## 1. 模拟退火: Simulated Annealing

# 1. 模拟退火: Simulated Annealing

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## Algorithm 2 模拟退火算法

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**Require:** 搜索半径  $r$ , 目标函数最大计算次数  $n_{\max}$ .

**Ensure:** 最小解  $\mathbf{x}^*$  和最小值  $f(\mathbf{x}^*)$ .

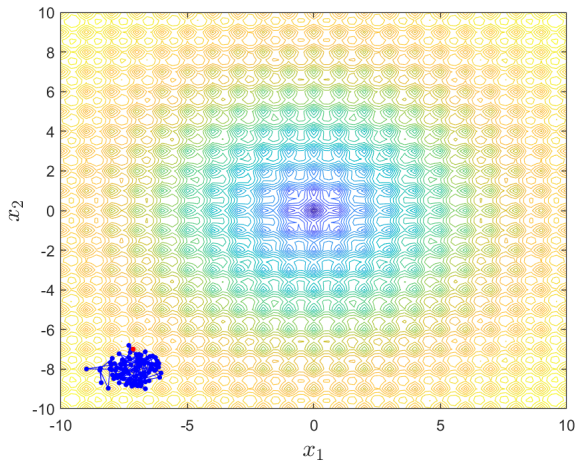
- 1: **for**  $i = 1$  to  $n_{\max} - 1$  **do**
- 2:   以  $\mathbf{x}^*$  为中心, 在  $r$  为半径的球内随机生成一个点  $\mathbf{x}^{(i)}$ .
- 3:   **if**  $f(\mathbf{x}^{(i)}) < f(\mathbf{x}^*)$  **then**
- 4:      $\mathbf{x}^* = \mathbf{x}^{(i)}$ .
- 5:   **else if**  $f(\mathbf{x}^{(i)}) \geq f(\mathbf{x}^*)$  **then**
- 6:     以概率

$$p = \frac{1}{1 + \exp\left(\frac{f(\mathbf{x}^{(i)}) - f(\mathbf{x}^*)}{0.9^i T_0}\right)}$$

接受新解  $\mathbf{x}^* = \mathbf{x}^{(i)}$ .

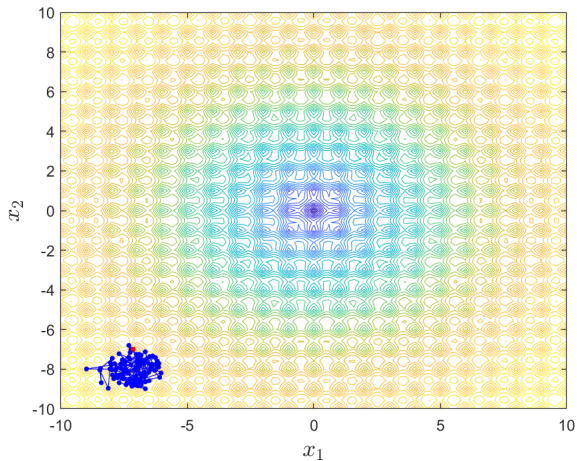
- 7:   **end if**
- 8: **end for**

# 1. 模拟退火：示例

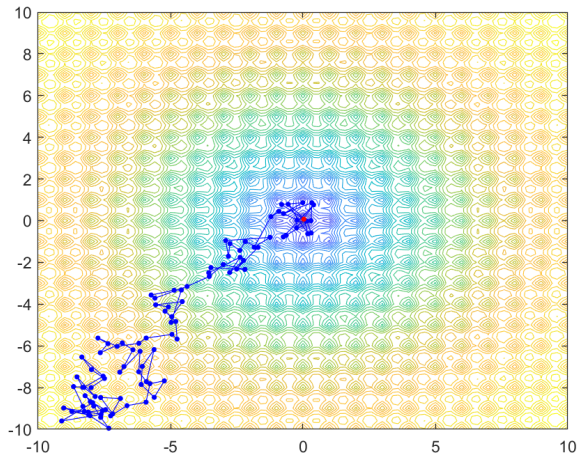


随机搜索算法

# 1. 模拟退火：示例



随机搜索算法



模拟退火算法

## 2. 遗传算法: Real-Coded Genetic Algorithm

模拟二进制交叉 (simulated binary crossover)

$$\beta = \begin{cases} (2)^{\frac{1}{\eta_c+1}}, & \text{if } u \leq 0.5 \\ \left(\frac{1}{2-2u}\right)^{\frac{1}{\eta_c+1}}, & \text{otherwise} \end{cases}$$

$$\mathbf{y}^{(1)} = 0.5 \left[ (\mathbf{x}^{(1)} + \mathbf{x}^{(2)}) - \beta (\mathbf{x}^{(2)} + \mathbf{x}^{(1)}) \right]$$

$$\mathbf{y}^{(2)} = 0.5 \left[ (\mathbf{x}^{(1)} + \mathbf{x}^{(2)}) + \beta (\mathbf{x}^{(2)} + \mathbf{x}^{(1)}) \right]$$

## 2. 遗传算法: Real-Coded Genetic Algorithm

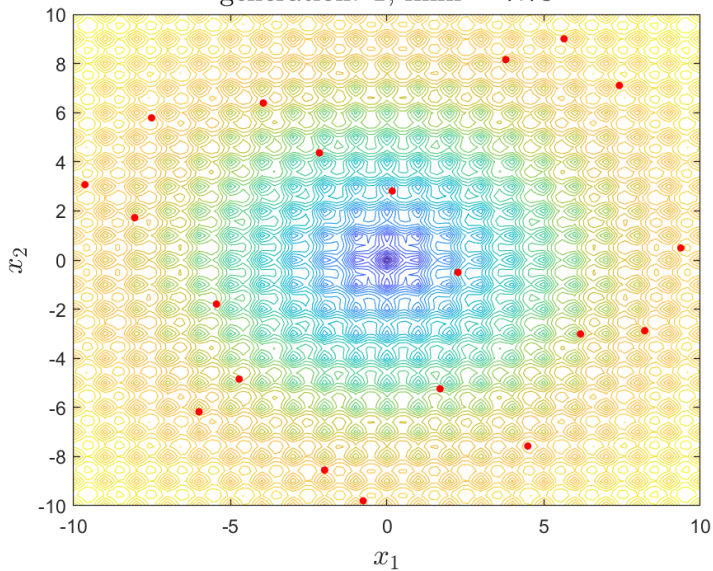
多项式变异 (polynomial mutation)

$$\delta = \begin{cases} (2)^{\frac{1}{\eta_m+1}}, & \text{if } u \leq 0.5 \\ 1 - (2 - 2u)^{\frac{1}{\eta_m+1}}, & \text{otherwise} \end{cases}$$

$$\mathbf{y} = \mathbf{x} + \delta(\mathbf{x}^l - \mathbf{x}^u)$$

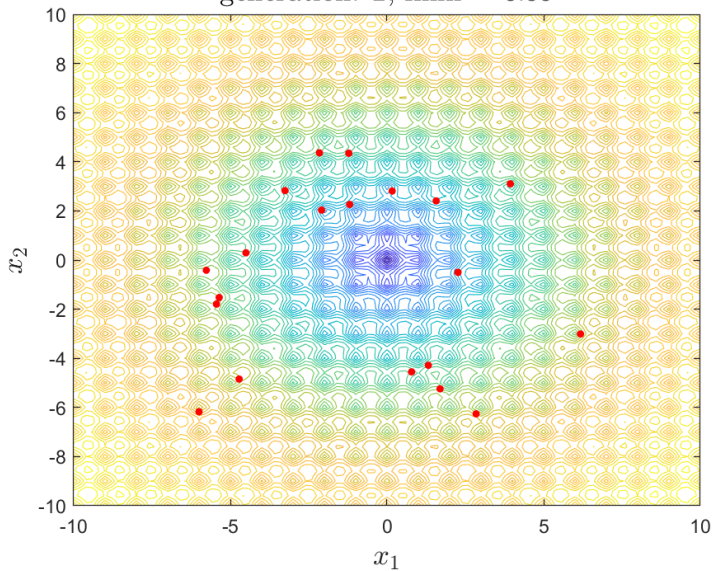
## 2. 遗传算法: Real-Coded Genetic Algorithm

generation: 1, fmin = 7.73



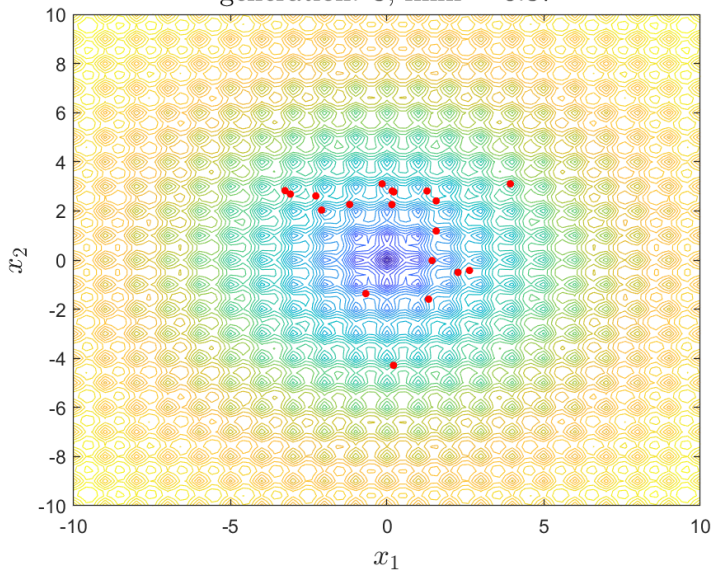
## 2. 遗传算法: Real-Coded Genetic Algorithm

generation: 2, fmin = 6.99



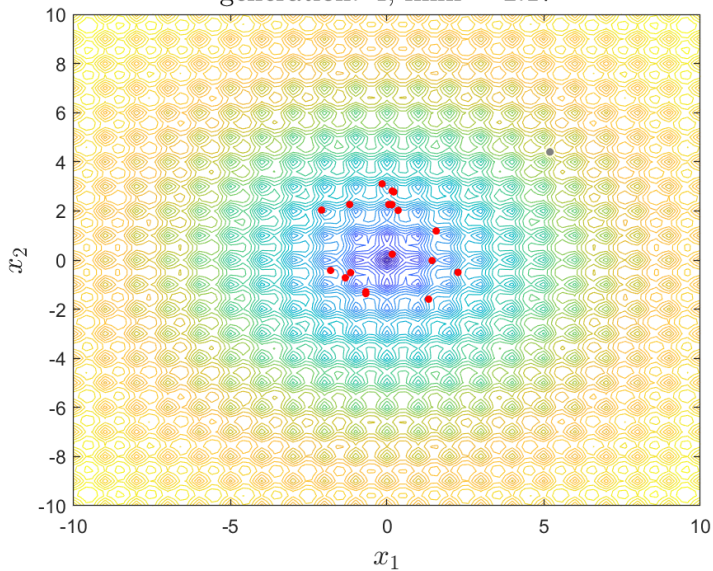
## 2. 遗传算法: Real-Coded Genetic Algorithm

generation: 3, fmin = 5.37



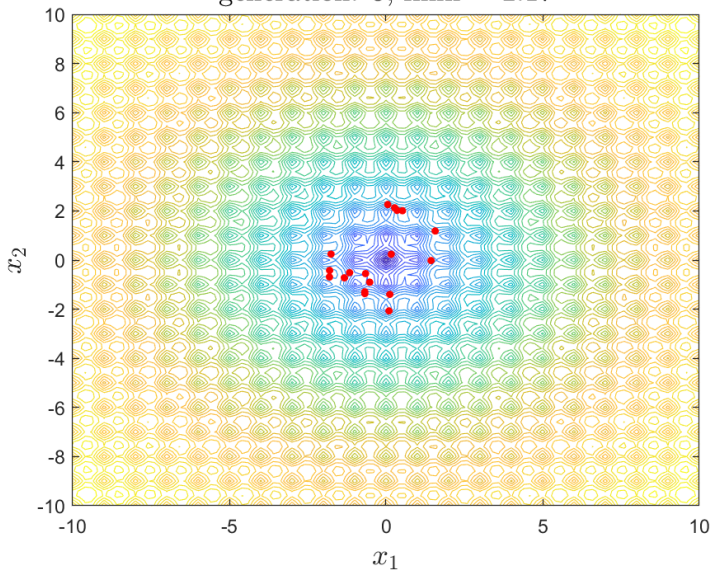
## 2. 遗传算法: Real-Coded Genetic Algorithm

generation: 4, fmin = 2.17



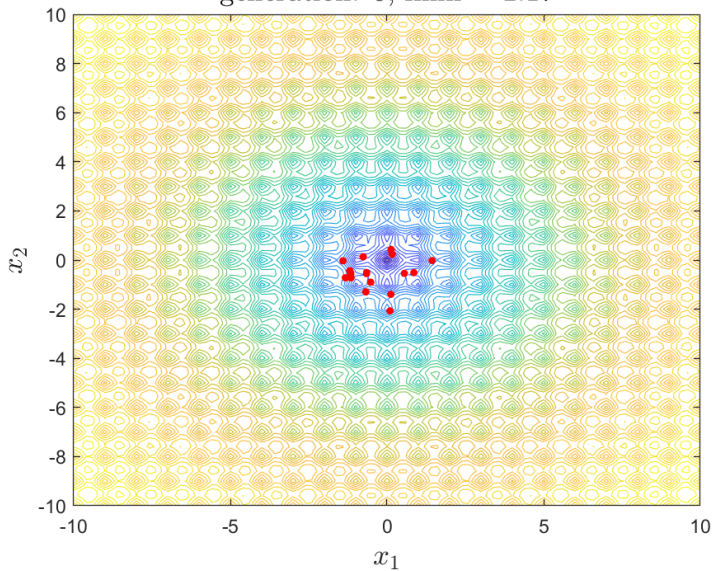
## 2. 遗传算法: Real-Coded Genetic Algorithm

generation: 5,  $f_{\min} = 2.17$



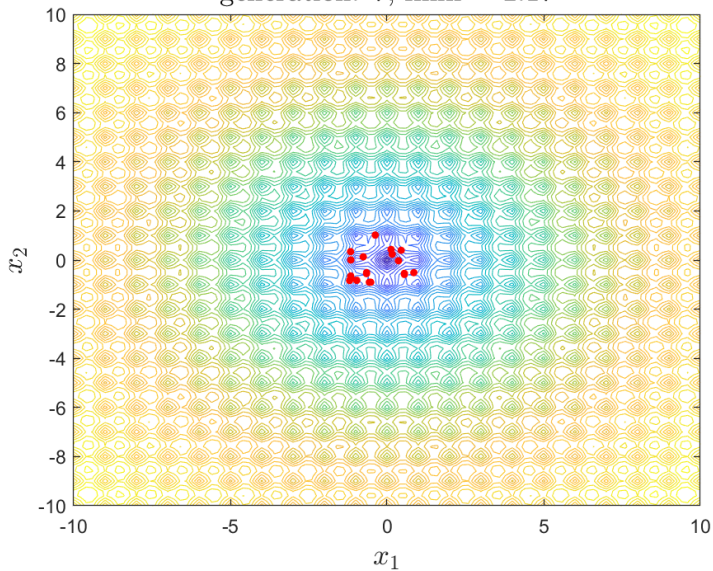
## 2. 遗传算法: Real-Coded Genetic Algorithm

generation: 6,  $f_{\min} = 2.17$



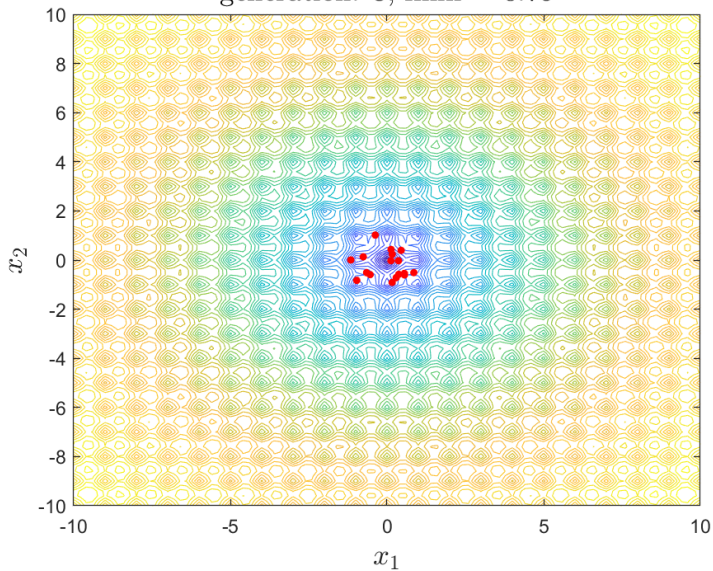
## 2. 遗传算法: Real-Coded Genetic Algorithm

generation: 7, fmin = 2.17



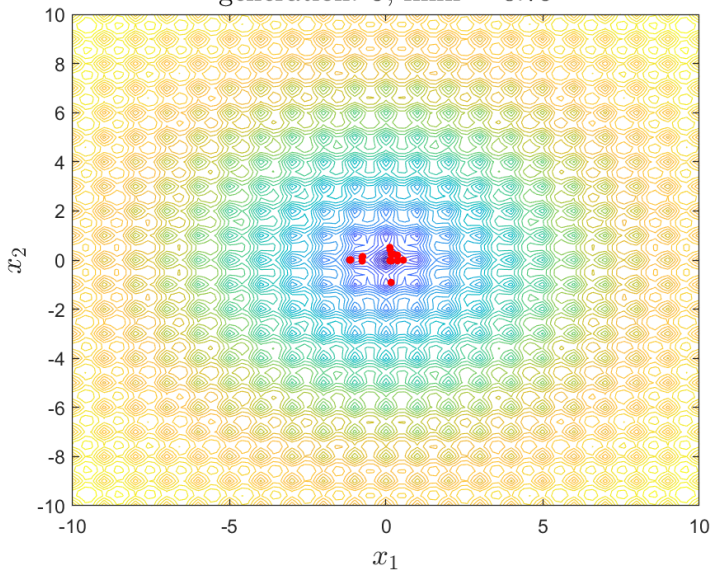
## 2. 遗传算法: Real-Coded Genetic Algorithm

generation: 8,  $f_{\min} = 0.75$



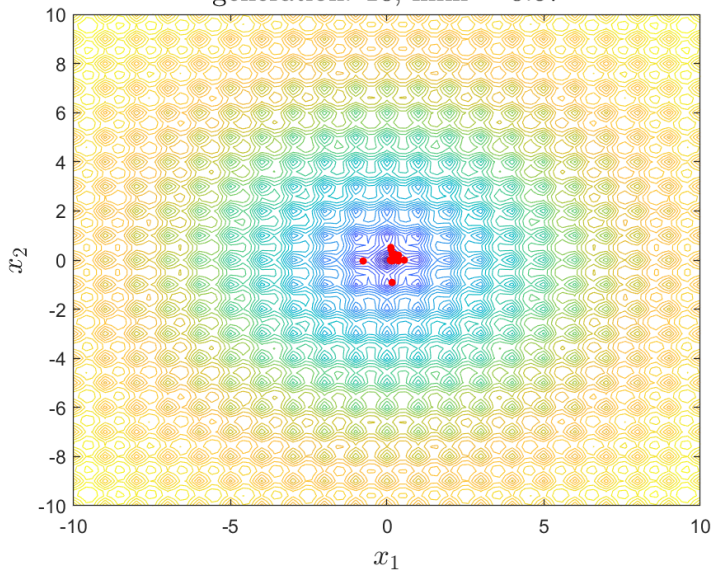
## 2. 遗传算法: Real-Coded Genetic Algorithm

generation: 9, fmin = 0.75



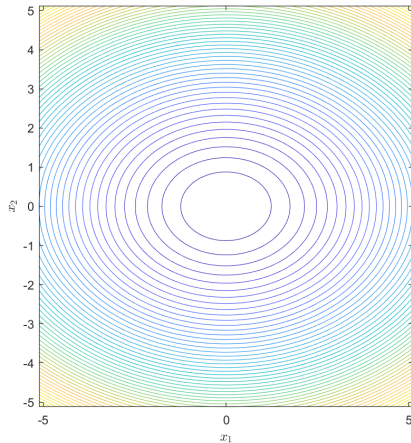
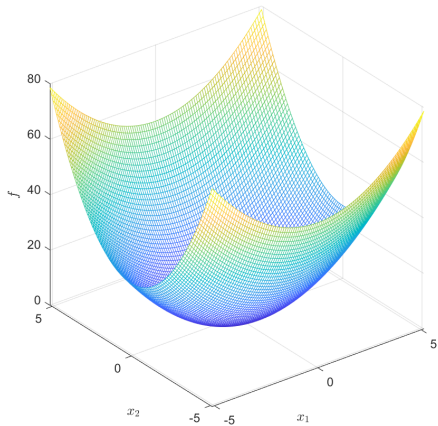
## 2. 遗传算法: Real-Coded Genetic Algorithm

generation: 10, fmin = 0.57

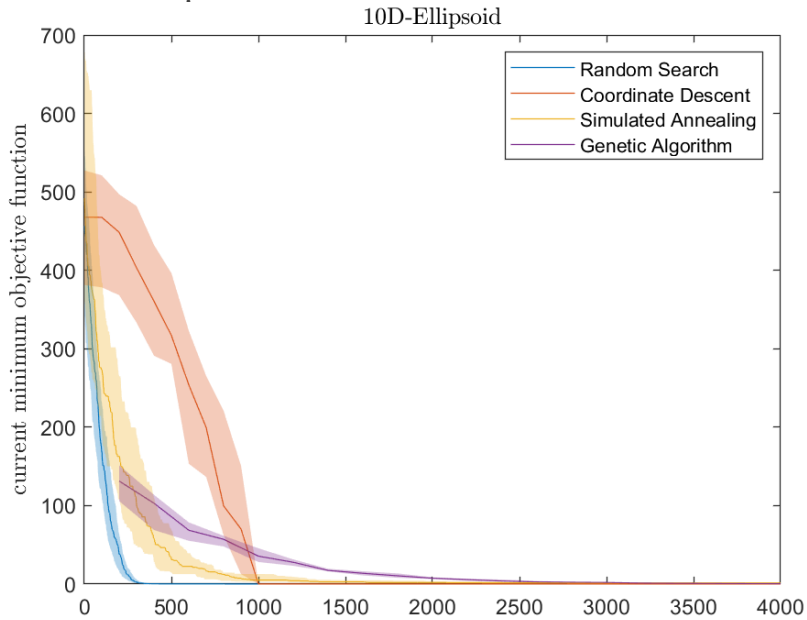


### 3. 零阶算法比较: Ellipsoid

$$f(\mathbf{x}) = \sum_{i=1}^d i x_i^2$$

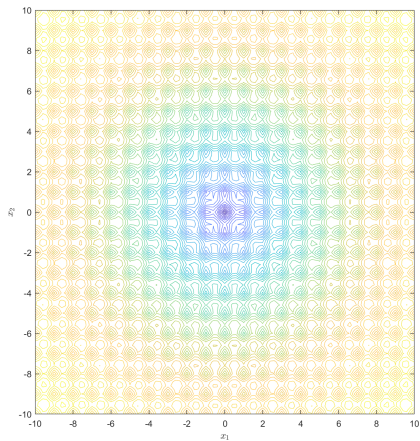
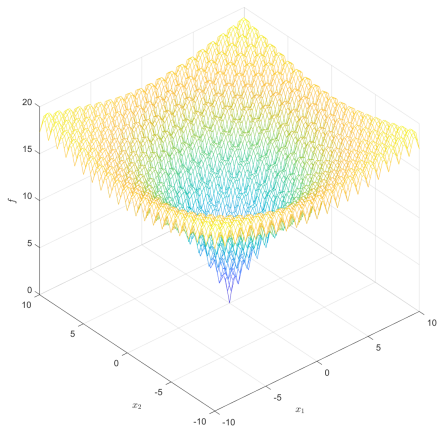


### 3. 零阶算法比较: Ellipsoid



### 3. 零阶算法比较: Ackley

$$f(\mathbf{x}) = -20 \exp \left( -0.2 \sqrt{\frac{1}{d} \sum_{i=1}^d x_i^2} \right) - \exp \left( \frac{1}{2} \sum_{i=1}^d \cos(2\pi x_i) \right) + 20 + \exp(1)$$



### 3. 零阶算法比较: Ackley

